

Alvin Lionel Thumi Thooko

Software Engineering student

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<https://lionel-portfolio-one.vercel.app/>

GitHub:

<https://github.com/AlvinLionel>

Summary

Software Engineering student with hands-on experience building full-stack applications and software systems using React, TypeScript, Node.js, Express, C, PHP and SQL. Experienced in developing REST APIs, integrating third-party services, working with relational and NoSQL databases, and building security-focused applications.

Tech stack

Languages: C, JavaScript, TypeScript, PHP, SQL

Frontend: HTML, CSS, React

Backend: Node.js, Express

Databases: MySQL, PostgreSQL, MongoDB

APIs & Integration: REST APIs, API Integration, Safaricom Daraja API

Security & Cryptography: Web Crypto, AES, RSA, ECDH/X25519

Tools: Git, GitHub, Vite, npm

Engineering: Requirements Engineering, Database management, Technical Documentation, Version Control

Projects

GitHub: https://github.com/AlvinLionel/My_Projects

LockBoxX- Browser-based security system

| *React, TypeScript, Web Crypto API*

Built a client-side security platform for encrypting and decrypting multiple resource types and with native-password support for some resources. Implements symmetric algorithms, public keys, key exchange algorithms and native-password protection techniques to secure resources.

<https://lockboxx.vercel.app/>

Full-Stack E-commerce Platform | Ongoing

| Node.js, Express, React, TypeScript, MySQL

Developing a full-stack e-commerce platform for discovering home products, nearby stores and home improvement services.

Implemented user authentication, product catalog, shopping cart functionality and REST API endpoints with an integrated interactive map functionality.

Checkout System | JavaScript, Node.js, MySQL, Bootstrap, Safaricom Daraja API

Built a store checkout page with product and transaction data managed through MySQL. Integrated Safaricom Daraja to support mobile payment and processing.

Static Server | Node.js

Built a custom static file server using Node.js for serving files over http.

Implemented event and error logging to monitor server activity and diagnose failures.

Weather Application | HTML, CSS, JavaScript, REST API

Developed a dynamic weather interface that extracts real-time data from an external API. Implemented responsive weather displays, cards and charts based on API responses and user requests

Education

The Co-operative University of Kenya
Bachelor of Science in Software Engineering
Expected Graduation: 2028.